Math 54 — Week 11

Solutions to the problem set

P1. Idea. Two distinct eigenvalues of a symmetric matrix give orthogonal eigenvectors; normalize them.

1. The trace is 7 and the determinant is $10 - 4 = 6$, so $\lambda^2 - 7\lambda + 6 = (\lambda - 6)(\lambda - 1)$: eigenvalues 6 and 1.
2. $\lambda = 6$: the first row of $A - 6I$ is $(-1, -2)$, so $x_1 = -2x_2$ and $(-2, 1)$ is an eigenvector. $\lambda = 1$: the perpendicular vector $(1, 2)$ (Thm. 1; check: $A(1, 2) = (1, 2)$).
3. Both have length $\sqrt{5}$, so
$$P = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 & 1 \\ 1 & 2 \end{bmatrix}, \qquad D = \begin{bmatrix} 6 & 0 \\ 0 & 1 \end{bmatrix}, \qquad A = PDP^{\mathsf{T}}.$$

P2. Idea. One simple eigenvalue and one double one; use Gram–Schmidt inside the double eigenspace.

1. Each row of A adds to 0, so $A(1, 1, 1) = \mathbf{0}$: eigenvalue 0 with eigenvector $(1, 1, 1)$.
2. $A - 3I$ has every entry -1, so it has rank 1, and $\lambda = 3$ has the 2-dimensional eigenspace $x_1 + x_2 + x_3 = 0$. The multiplicities 1 and 2 account for all three eigenvalues.
3. A basis of that eigenspace is $(-1, 1, 0)$, $(-1, 0, 1)$. Gram–Schmidt:
$$(-1, 0, 1) - \frac{1}{2}(-1, 1, 0) = \left(-\frac{1}{2}, -\frac{1}{2}, 1\right), \quad \text{use } (-1, -1, 2).$$
4. Normalize:
$$P = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 1/\sqrt{2} & -1/\sqrt{6} \\ 1/\sqrt{3} & 0 & 2/\sqrt{6} \end{bmatrix}, \qquad D = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{bmatrix}, \qquad A = PDP^{\mathsf{T}}.$$

P3. Idea. $A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^{\mathsf{T}} + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^{\mathsf{T}}$ with the unit eigenvectors from P1.

1. With $\mathbf{u}_1 = \frac{1}{\sqrt{5}}(-2, 1)$ and $\mathbf{u}_2 = \frac{1}{\sqrt{5}}(1, 2)$,
$$\mathbf{u}_1 \mathbf{u}_1^{\mathsf{T}} = \frac{1}{5} \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix}, \qquad \mathbf{u}_2 \mathbf{u}_2^{\mathsf{T}} = \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$
2. Check the sum:
$$6 \cdot \frac{1}{5} \begin{bmatrix} 4 & -2 \\ -2 & 1 \end{bmatrix} + 1 \cdot \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \frac{1}{5} \begin{bmatrix} 25 & -10 \\ -10 & 10 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ -2 & 2 \end{bmatrix}.$$

P4. Idea. For symmetric A, $A^2 = A^{\mathsf{T}} A$, and $\mathbf{x}^{\mathsf{T}} A^{\mathsf{T}} A \mathbf{x} = \|A\mathbf{x}\|^2$.

1. Since $A^{\mathsf{T}} = A$, for every $\mathbf{x}$,
$$\|A\mathbf{x}\|^2 = \mathbf{x}^{\mathsf{T}} A^{\mathsf{T}} A \mathbf{x} = \mathbf{x}^{\mathsf{T}} A^2 \mathbf{x} = 0.$$
2. So $A\mathbf{x} = \mathbf{0}$ for every $\mathbf{x}$; taking $\mathbf{x} = \mathbf{e}_j$ shows every column of A is $\mathbf{0}$, so $A = 0$.

P5. Idea. Eigenvectors of $A^\mathsf{T}A$ give V; then $\mathbf{u}_i = A\mathbf{v}_i/\sigma_i$.

1. The matrix
$$A^\mathsf{T}A = \begin{bmatrix} 5 & -3 \\ -3 & 5 \end{bmatrix}$$
has characteristic polynomial $\lambda^2 - 10\lambda + 16 = (\lambda - 8)(\lambda - 2)$. For 8, the first row of $A^\mathsf{T}A - 8I$ is $(-3, -3)$, so $\mathbf{v}_1 = (1, -1)/\sqrt{2}$; for 2, the perpendicular $\mathbf{v}_2 = (1, 1)/\sqrt{2}$. So $\sigma_1 = 2\sqrt{2}$ and $\sigma_2 = \sqrt{2}$.
2. $A\mathbf{v}_1 = (0, -4)/\sqrt{2}$, so $\mathbf{u}_1 = (0, -1)$; $A\mathbf{v}_2 = (2, 0)/\sqrt{2}$, so $\mathbf{u}_2 = (1, 0)$.
3. So
$$A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 2\sqrt{2} & 0 \\ 0 & \sqrt{2} \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}.$$
Check: $U\Sigma$ has rows $(0, \sqrt{2})$ and $(-2\sqrt{2}, 0)$, and multiplying by V^T gives the rows $(1, 1)$ and $(-2, 2)$ of A.

P6. Idea. Rank 1: one nonzero singular value. The other columns of U form an orthonormal basis of $\operatorname{Nul} A^\mathsf{T}$.

1. The matrix
$$A^\mathsf{T}A = \begin{bmatrix} 9 & 9 \\ 9 & 9 \end{bmatrix}$$
has eigenvalues 18 and 0 (trace 18, determinant 0), with unit eigenvectors $\mathbf{v}_1 = (1, 1)/\sqrt{2}$ and $\mathbf{v}_2 = (1, -1)/\sqrt{2}$. So $\sigma_1 = \sqrt{18} = 3\sqrt{2}$, $\sigma_2 = 0$, and $\operatorname{rank} A = 1$.
2. $A\mathbf{v}_1 = (4, 4, 2)/\sqrt{2}$, of length $6/\sqrt{2} = 3\sqrt{2}$, so $\mathbf{u}_1 = \frac{1}{3}(2, 2, 1)$.
3. $\operatorname{Nul} A^\mathsf{T}$ is the plane $2x_1 + 2x_2 + x_3 = 0$. It contains $(1, -1, 0)$. A second vector in the plane orthogonal to $(1, -1, 0)$ satisfies $2x_1 + 2x_2 + x_3 = 0$ and $x_1 - x_2 = 0$; with $x_1 = x_2 = 1$ this gives $(1, 1, -4)$. So $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1, 0)$ and $\mathbf{u}_3 = \frac{1}{3\sqrt{2}}(1, 1, -4)$.
4. So $A = U\Sigma V^\mathsf{T}$ with
$$U = \begin{bmatrix} 2/3 & 1/\sqrt{2} & 1/(3\sqrt{2}) \\ 2/3 & -1/\sqrt{2} & 1/(3\sqrt{2}) \\ 1/3 & 0 & -4/(3\sqrt{2}) \end{bmatrix}, \qquad \Sigma = \begin{bmatrix} 3\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad V = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$
5. Bases: $\operatorname{Col} A$: $\mathbf{u}_1$; $\operatorname{Nul} A^\mathsf{T}$: $\mathbf{u}_2, \mathbf{u}_3$; $\operatorname{Row} A$: $\mathbf{v}_1$; $\operatorname{Nul} A$: $\mathbf{v}_2$ (check: $A(1, -1) = \mathbf{0}$).

P7. Idea. Distinct real roots; two conditions at $t = 0$ give a 2×2 system.

1. $r^2 - 3r - 10 = (r - 5)(r + 2)$, so $y = c_1 e^{5t} + c_2 e^{-2t}$.
2. $y(0) = c_1 + c_2 = 0$ and $y'(0) = 5c_1 - 2c_2 = 7$. Substituting $c_2 = -c_1$ gives $7c_1 = 7$, so $c_1 = 1$ and $c_2 = -1$.
3. $y = e^{5t} - e^{-2t}$.

P8. Idea. A repeated root r gives e^{rt} and te^{rt}; then $c_1 = y(0)$, $c_2 = y'(0) - ry(0)$.

1. $4r^2 + 12r + 9 = (2r + 3)^2$, so $r = -\frac{3}{2}$ is a double root and $y = (c_1 + c_2 t)e^{-3t/2}$.
2. $c_1 = y(0) = 2$ and $c_2 = y'(0) - r\,y(0) = 0 + \frac{3}{2} \cdot 2 = 3$.
3. $y = (2 + 3t)e^{-3t/2}$.

P9. Idea. Complex roots $\alpha \pm i\beta$ give $e^{\alpha t}(c_1 \cos \beta t + c_2 \sin \beta t)$.

1. $r^2 + 2r + 10 = 0$ gives $\alpha = -1$, and $4ac - b^2 = 36$ gives $\beta = 6/2 = 3$. So $y = e^{-t}(c_1 \cos 3t + c_2 \sin 3t)$.
2. $c_1 = y(0) = 0$, and $c_2 = (y'(0) - \alpha y(0))/\beta = 6/3 = 2$.
3. $y = 2e^{-t} \sin 3t$. Check: $y' = 2e^{-t}(3 \cos 3t - \sin 3t)$, so $y'(0) = 6$.

P10. Idea. Differentiate with the product rule; $\cos^2 + \sin^2 = 1$ collects the terms.

1. $y_1 = e^{\alpha t}\cos\beta t$ has $y_1' = e^{\alpha t}(\alpha\cos\beta t - \beta\sin\beta t)$, and $y_2 = e^{\alpha t}\sin\beta t$ has $y_2' = e^{\alpha t}(\alpha\sin\beta t + \beta\cos\beta t)$.
2. So

$$\begin{aligned} W = y_1 y_2' - y_1' y_2 &= e^{2\alpha t}\big[\cos\beta t\,(\alpha\sin\beta t + \beta\cos\beta t) - \sin\beta t\,(\alpha\cos\beta t - \beta\sin\beta t)\big] \\ &= e^{2\alpha t}\big[\beta\cos^2\beta t + \beta\sin^2\beta t\big] = \beta e^{2\alpha t}, \end{aligned}$$

   because the two $\alpha\sin\beta t\cos\beta t$ terms cancel.
3. If two functions are dependent, say $y_2 = ky_1$, their Wronskian is $y_1\,ky_1' - y_1'\,ky_1 = 0$ everywhere (the case $y_1 = ky_2$ is the same). Here $W = \beta e^{2\alpha t} \neq 0$ when $\beta \neq 0$, so y_1 and y_2 are independent.